

## APPLICATION #2: SELF-BOOTSTRAP TOOLCHAIN

12 Patent Drawings -- ASCII Reference Diagrams

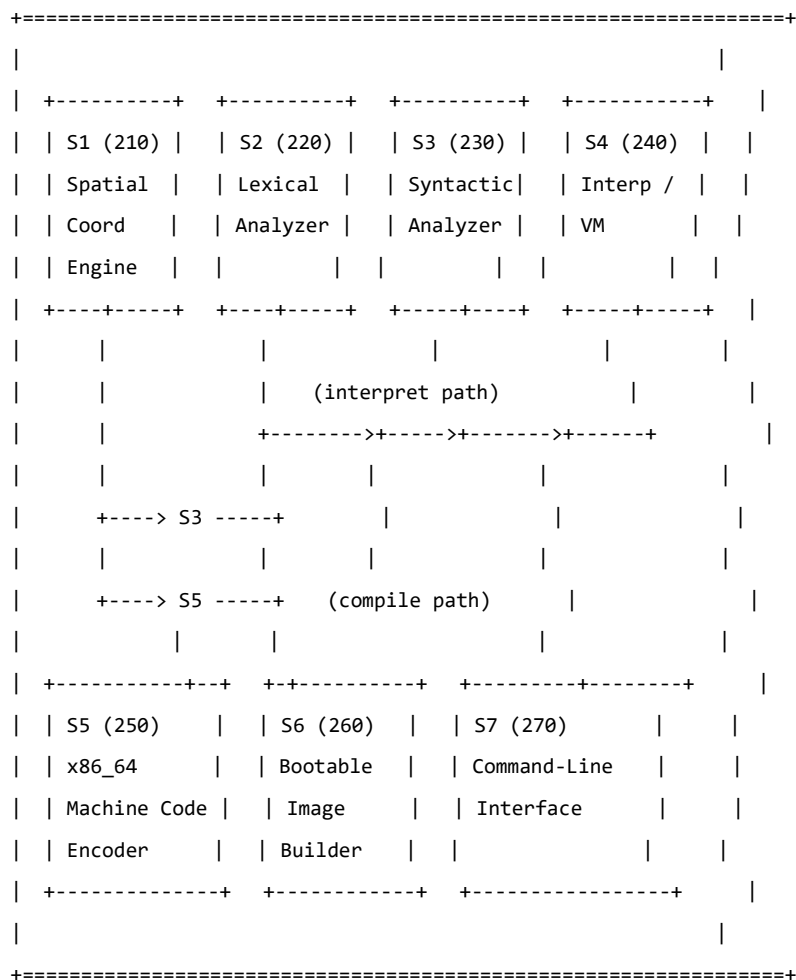
for tracing with Draw.io / Visio / PowerPoint

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FIG. 1 -- Toolchain Subsystem Block Diagram

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hl-bootstrap.hl (4,572 lines, 208 functions)



Data flow:

Interpret: S7 --> S2 --> S3 --> S4 --> (result)

Compile: S7 --> S2 --> S3 --> S5 --> S6 --> (bootable image)

S1 connected to S3 and S5 (coordinate primitives)

FIG. 2 -- Self-Bootstrap Cycle

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hl-bootstrap.hl  
(Source Code)

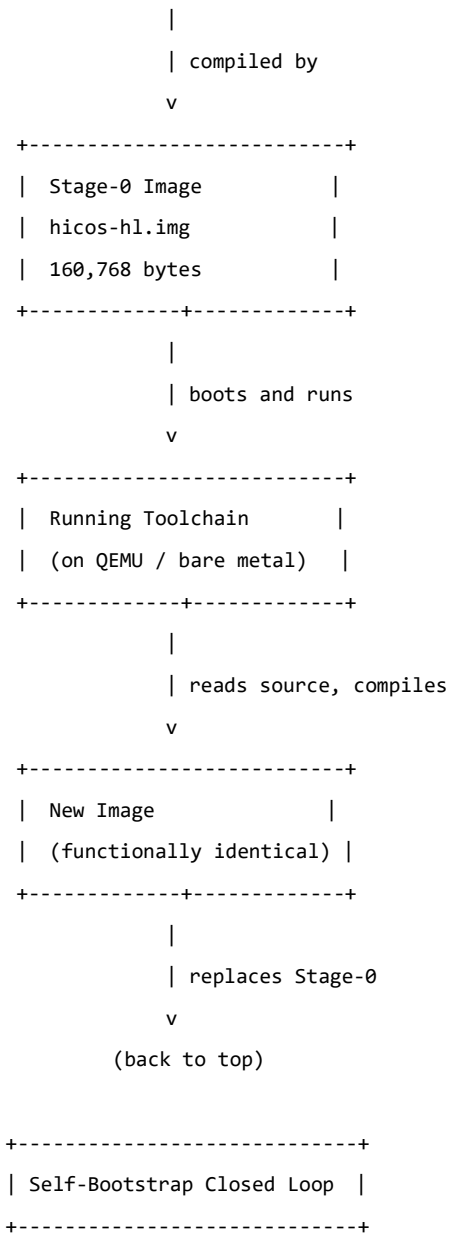
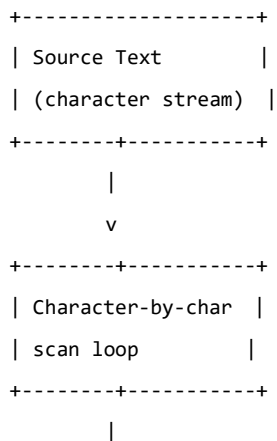


FIG. 3 -- Lexical Analysis Pipeline

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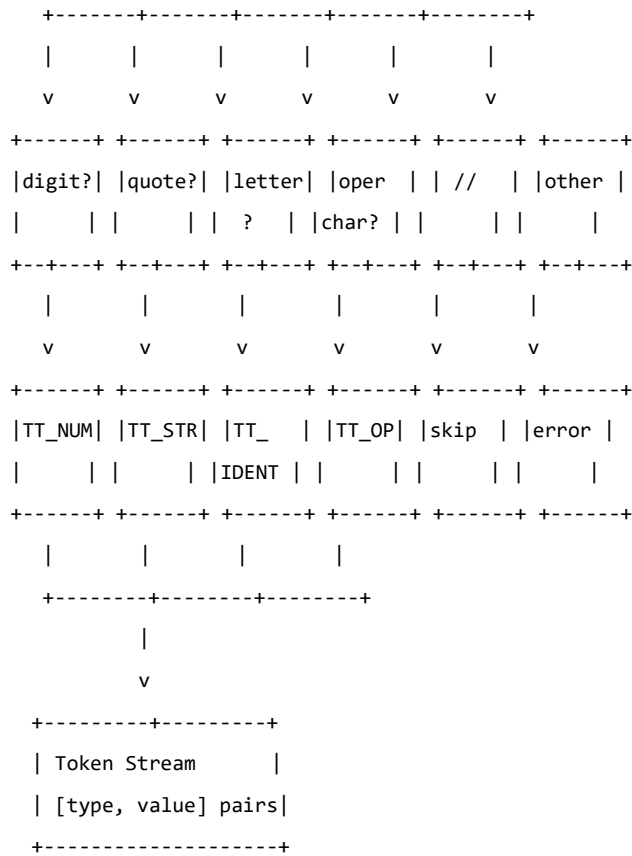
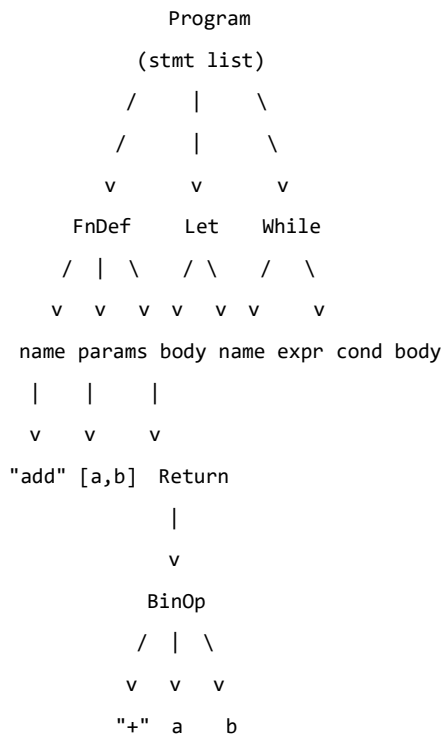


FIG. 4 -- Abstract Syntax Tree Structure

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Node types:

```

FnDef    -> [name, params, body]
If       -> [cond, then, else]
While    -> [cond, body]
Let       -> [name, expr]
BinOp    -> [op, left, right]
FnCall   -> [name, args]
* Quadrant -> [name, body] <-- UNIQUE to H-L

```

Example: `fn add(a, b) { return a + b; }`

FIG. 5 -- Interpreter/VM Block Diagram

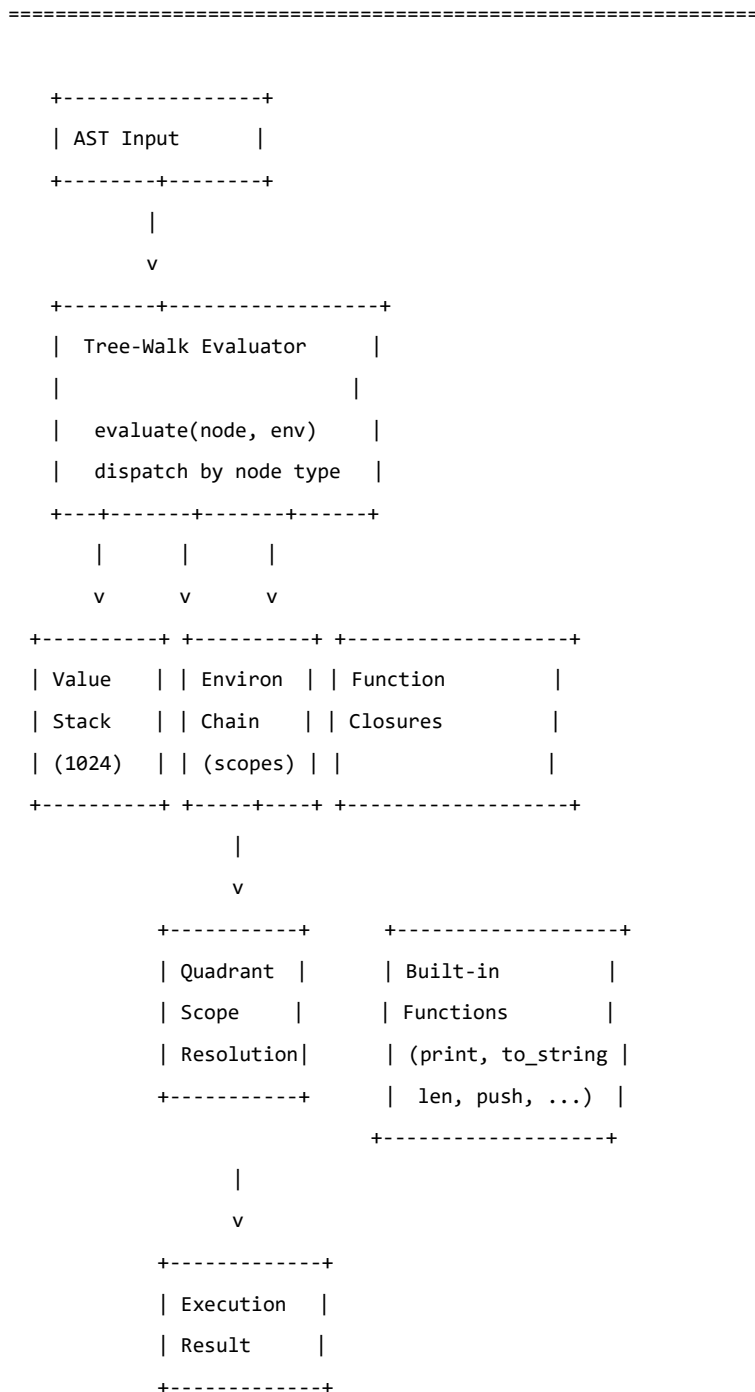


FIG. 6 -- x86\_64 Machine Code Encoding

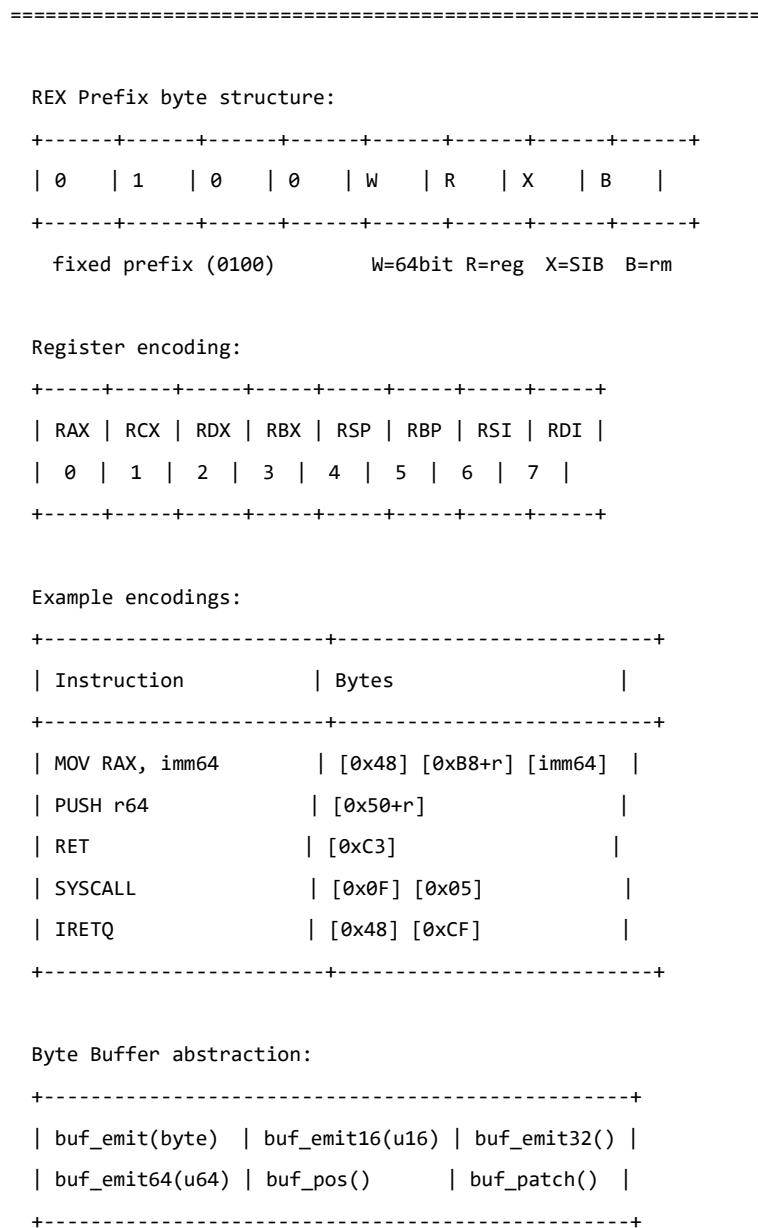
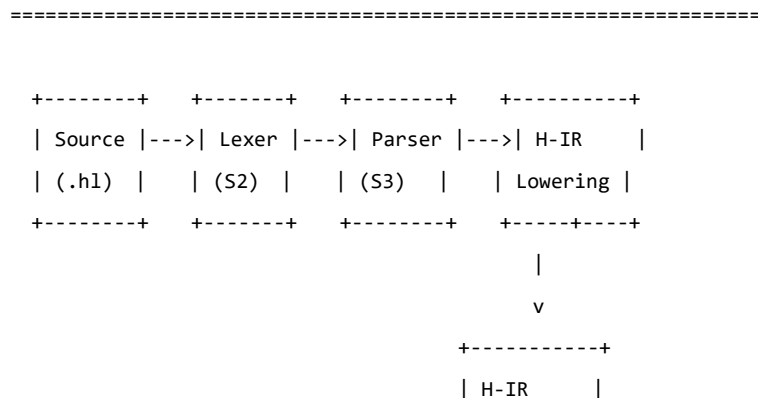


FIG. 7 -- Multi-Stage Compilation Pipeline



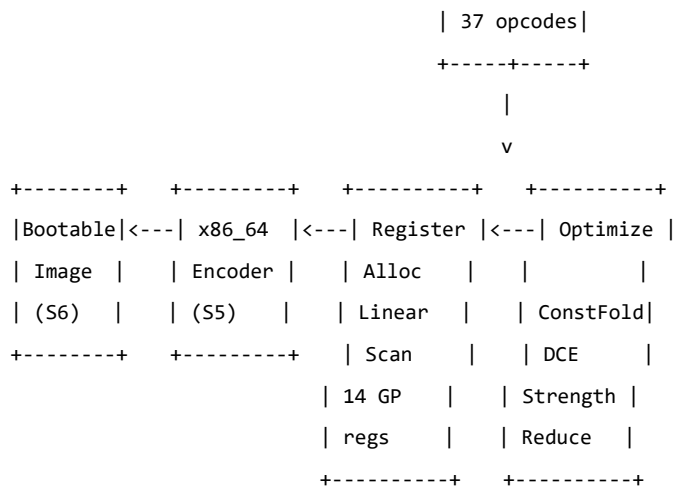


FIG. 8 -- Boot Image Disk Layout

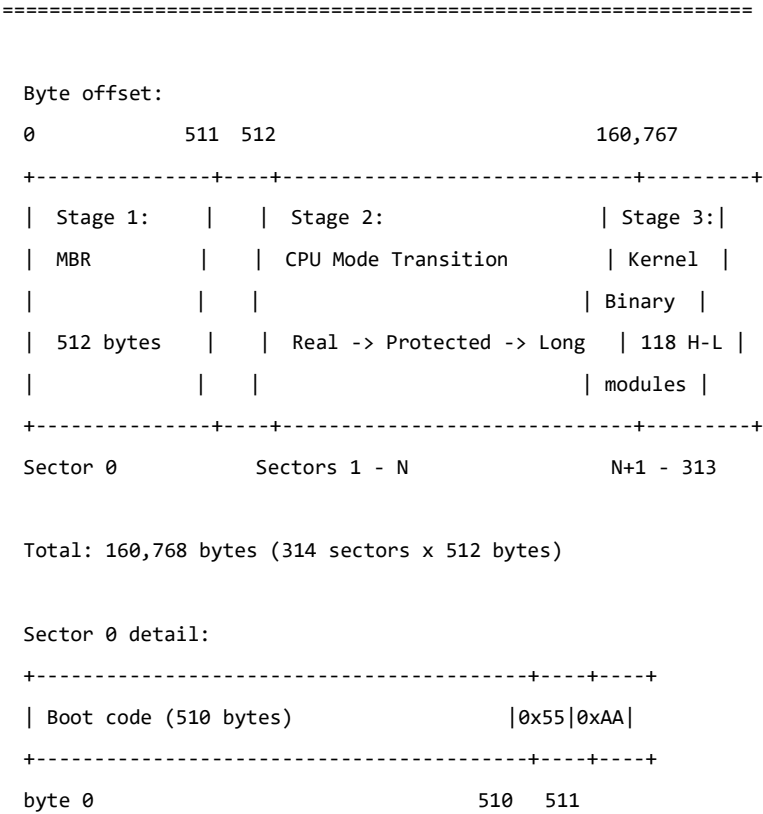
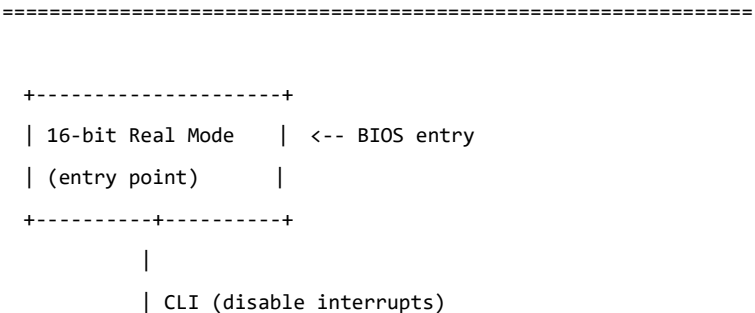


FIG. 9 -- CPU Mode Transition Sequence



```

    | LGDT (load GDT)
    | Set CR0.PE = 1
    | Far JMP selector 8
v
+-----+
| 32-bit Protected |
| Mode            |
+-----+-----+
|
| Copy kernel to 0x100000 (REP MOVSD)
| Build page tables:
|   PML4 @ 0x1000
|   PDPT @ 0x2000
|   PD   @ 0x3000
| Set CR3 = 0x1000 (PML4)
| Set CR4.PAE = 1
| Set EFER.LME = 1 (MSR 0xC0000080)
| Set CR0.PG = 1
| Far JMP selector 24
v
+-----+
| 64-bit Long Mode | <-- target mode
| (target)         |
+-----+-----+
|
| Set RSP = 0x70000
| JMP kernel_entry
v
[ kernel runs ]

```

FIG. 10 -- H-IR Instruction Set (37 opcodes)

| =====                           |            |                 |                |  |
|---------------------------------|------------|-----------------|----------------|--|
| #                               | Mnemonic   | Format          | Description    |  |
| +-----+-----+-----+-----+-----+ |            |                 |                |  |
|                                 | ARITHMETIC |                 |                |  |
| +-----+-----+-----+-----+-----+ |            |                 |                |  |
| 1                               | IR_CONST   | dst, imm        | Load constant  |  |
| 2                               | IR_ADD     | dst, src1, src2 | Addition       |  |
| 3                               | IR_SUB     | dst, src1, src2 | Subtraction    |  |
| 4                               | IR_MUL     | dst, src1, src2 | Multiplication |  |
| 5                               | IR_DIV     | dst, src1, src2 | Division       |  |
| 6                               | IR_MOD     | dst, src1, src2 | Modulo         |  |
| 7                               | IR_NEG     | dst, src        | Negate         |  |

|                                       |  |              |                 |                    |  |  |
|---------------------------------------|--|--------------|-----------------|--------------------|--|--|
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | BITWISE      |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 8                                     |  | IR_AND       | dst, src1, src2 | Bitwise AND        |  |  |
| 9                                     |  | IR_OR        | dst, src1, src2 | Bitwise OR         |  |  |
| 10                                    |  | IR_XOR       | dst, src1, src2 | Bitwise XOR        |  |  |
| 11                                    |  | IR_SHL       | dst, src1, src2 | Shift left         |  |  |
| 12                                    |  | IR_SHR       | dst, src1, src2 | Shift right        |  |  |
| 13                                    |  | IR_NOT       | dst, src        | Bitwise NOT        |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | COMPARISON   |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 14                                    |  | IR_EQ        | dst, src1, src2 | Equal              |  |  |
| 15                                    |  | IR_NE        | dst, src1, src2 | Not equal          |  |  |
| 16                                    |  | IR_LT        | dst, src1, src2 | Less than          |  |  |
| 17                                    |  | IR_LE        | dst, src1, src2 | Less or equal      |  |  |
| 18                                    |  | IR_GT        | dst, src1, src2 | Greater than       |  |  |
| 19                                    |  | IR_GE        | dst, src1, src2 | Greater or equal   |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | CONTROL FLOW |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 20                                    |  | IR_JMP       | label           | Unconditional jump |  |  |
| 21                                    |  | IR_JZ        | src, label      | Jump if zero       |  |  |
| 22                                    |  | IR_JNZ       | src, label      | Jump if not zero   |  |  |
| 23                                    |  | IR_LABEL     | id              | Label definition   |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | FUNCTION     |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 24                                    |  | IR_CALL      | dst, fn, args   | Function call      |  |  |
| 25                                    |  | IR_RET       | src             | Return value       |  |  |
| 26                                    |  | IR_PARAM     | dst, index      | Get parameter      |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | MEMORY       |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 27                                    |  | IR_LOAD      | dst, addr       | Load from memory   |  |  |
| 28                                    |  | IR_STORE     | addr, src       | Store to memory    |  |  |
| 29                                    |  | IR_ALLOCA    | dst, size       | Stack allocation   |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | SSA          |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 30                                    |  | IR_PHI       | dst, src1, src2 | SSA phi function   |  |  |
| 31                                    |  | IR_MOVE      | dst, src        | Register move      |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
|                                       |  | EXTENDED     |                 |                    |  |  |
| +-----+-----+-----+-----+-----+-----+ |  |              |                 |                    |  |  |
| 32                                    |  | IR_SYSCALL   | dst, nr, args   | System call        |  |  |



|                                 |            |                |                        |
|---------------------------------|------------|----------------|------------------------|
| 33                              | IR_LEA     | dst, base, off | Load effective address |
| 34                              | IR_ZERO    | dst            | Zero register          |
| 35                              | IR_SIGN_EX | dst, src       | Sign extend            |
| 36                              | IR_CMov    | dst, cond, src | Conditional move       |
| 37                              | IR_MEMCPY  | dst, src, len  | Memory copy            |
| +-----+-----+-----+-----+-----+ |            |                |                        |

FIG. 11 -- Register Allocation Strategy

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#### 14 General Purpose Registers (allocatable):

|                                 |                         |              |  |
|---------------------------------|-------------------------|--------------|--|
| +-----+-----+-----+-----+-----+ |                         |              |  |
| RAX                             | ====[live range A]===== |              |  |
| RCX                             | ====[live range B]===== |              |  |
| RDX                             | ==[C]==                 | ====[D]====  |  |
| RBX                             |                         | ====[E]===== |  |
| RSI                             | ====[F]===              |              |  |
| RDI                             |                         | ====[G]====  |  |
| R8                              |                         | ====[H]===== |  |
| R9                              | ==[I]==                 |              |  |
| R10                             |                         | ====[J]===== |  |
| R11                             |                         | ====[K]====  |  |
| R12                             |                         | ==[L]===     |  |
| R13                             | ====[M]=====            |              |  |
| R14                             |                         | ====[N]====  |  |
| R15                             |                         | ====[O]====  |  |
| +-----+-----+-----+-----+-----+ |                         |              |  |

#### Reserved (NOT allocatable):

|                                 |                                                      |  |           |
|---------------------------------|------------------------------------------------------|--|-----------|
| +-----+-----+-----+-----+-----+ |                                                      |  |           |
| RSP                             | XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |  | Stack ptr |
| RBP                             | XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |  | Frame ptr |
| +-----+-----+-----+-----+-----+ |                                                      |  |           |

#### Spill to stack:

|            |                                                |
|------------|------------------------------------------------|
| +-----+    |                                                |
| [RBP - 8 ] | var1                                           |
| [RBP - 16] | var2  <-- "Spill to stack" for exceeded ranges |
| [RBP - 24] | var3                                           |
| +-----+    |                                                |

#### System V AMD64 ABI calling convention:

|                                       |      |      |      |      |      |        |
|---------------------------------------|------|------|------|------|------|--------|
| +-----+-----+-----+-----+-----+-----+ |      |      |      |      |      |        |
| arg0                                  | arg1 | arg2 | arg3 | arg4 | arg5 | return |
| RDI                                   | RSI  | RDX  | RCX  | R8   | R9   | RAX    |

```
+-----+-----+-----+-----+-----+-----+-----+
```

FIG. 12 -- OS Toolchain Dependency Comparison

```
=====
```

| OS       | Kernel   | Compiler | Assembler | Linker   | Image    | Ext   |
|----------|----------|----------|-----------|----------|----------|-------|
|          | Language |          |           |          | Tool     | Deps  |
| Linux    | C + asm  | GCC (C)  | gas (C)   | ld (C)   | make+dd  | 4+    |
| Redox    | Rust     | rustc    | LLVM      | lld      | cargo    | 3+    |
|          |          | (Rust)   | (C++)     |          |          |       |
| TempleOS | HolyC    | HolyC    | built-in  | built-in | built-in | 1     |
|          | + asm    | + asm    |           |          |          | (asm) |
| Oberon   | Oberon   | Oberon   | built-in  | built-in | external | 1     |
| *HicOS*  | *H-L*    | *H-L*    | *H-L*     | *H-L*    | *H-L*    | * 0*  |

```
+-----+-----+-----+-----+-----+-----+-----+
```

^^^^^^^

HicOS row: ZERO

external dependencies